

Data types, variables, constants and input/output ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- List C data types and format specifiers
- Use printf, scanf, getchar, getch, puts, gets (exam style)
- Convert algebraic expressions into C

Type	Typical size (Turbo C 16-bit)	Specifier	Example
char	1 byte	%c	grade = 'A'
int	2 bytes	%d	n = 25
float	4 bytes	%f	avg = 72.5
double	8 bytes	%lf	pi = 3.14159
long	4 bytes	%ld	pop = 16000000L

Constants: 25 integer, 3.14 float, 'A' character, "BIEK" string. Variables must be declared before use: int roll; float marks;

```
int a, b, sum;
printf("Enter two numbers: ");
scanf("%d%d", &a, &b); /* & is the address operator */
sum = a + b;
printf("Sum = %d\n", sum);
```

Function	Use
printf	Formatted output to screen
scanf	Formatted input; needs & for scalars
getchar	Read one character (waits for Enter)
getch	Read one character without echo (conio.h)
gets / puts	String line I/O (unsafe in modern C, still in old papers)

Algebra → C (every year)

Mathematics	C expression
$A = \frac{1}{2} \times \text{base} \times \text{height}$	<code>A = 0.5 * base * height;</code>
$x = 3ab^3 + 3a^2b$	<code>x = 3*a*b*b*b + 3*a*a*b;</code>
$d = b^2 - 4ac $	<code>d = fabs(b*b - 4*a*c); /* math.h */</code>
$A = \pi r^2$	<code>A = 3.14 * r * r;</code>
$(a+b)^2 / ab$	<code>((a+b)*(a+b))/(a*b)</code>
$y = (a+b)^n$	<code>y = pow(a+b, n);</code>

WORKED EXAMPLE / BOARD STYLE

If $a=2$, $b=5$, $c=10$:

$200*((a*10)+c)$ wait — follow the paper's parentheses exactly.

$a*b*10/c+a+c = 2*5*10/10+2+10 = 100/10+2+10 = 10+2+10 = 22$ (integer division if all int).

$a-(c/b)/2+100 = 2-(10/5)/2+100 = 2-2/2+100 = 2-1+100 = 101$ with int division of $10/5$ first.

COMMON BOARD MISTAKE

`scanf("%d", &n)` — forgetting `&` is the #1 runtime bug. `%` with float is undefined; `%` is for integers only.

اردو میں سمجھیں

int عدد صحیح، float اعشاری، char ایک حرف، `%%`، `%f`، `%c`، `%d` فارمیٹ۔ `scanf` میں `&` پتہ دیتا ہے۔
ریاضی کو C میں `*` / اور `pow()` سے لکھیں۔ $\frac{1}{2}$ کو `*0.5` بنائیں۔

CHECK YOURSELF

1. Format specifier for int?

2. Why `&` in `scanf`?

Answers: 1. `%d` · 2. Pass the address so the value can be stored in the variable

MCQs

1. `%` operator is only used with:

- a. float
- b. int
- c. double
- d. string

2. The address operator is:

- a. `*`
- b. `&`
- c. `%`
- d. `#`

Key: 1=b, 2=b

Short questions (3 marks each)

1. How does `scanf()` differ from `getchar()`?
2. What is the address operator? Explain with example.
3. Convert $A = \frac{1}{2} \text{ base} \cdot \text{height}$ and $A = \pi r^2$ into C.
4. Define increment and decrement operators.

Model short answers

1. scanf reads formatted data (numbers, strings) using specifiers; getchar reads a single character from the keyboard buffer.
2. & gives the memory address. `int x=5; printf("%u", &x);` prints the address of x. `scanf("%d", &x)` stores input at that address.
3. `A=0.5*base*height;` `A=3.14*r*r;` (or `PI*r*r`).
4. ++ adds 1, -- subtracts 1. Pre (++k) changes then uses; post (k++) uses then changes.

Long questions (board Section C)

1. What are different types of operators in C? Explain with examples. Also discuss printf and scanf.